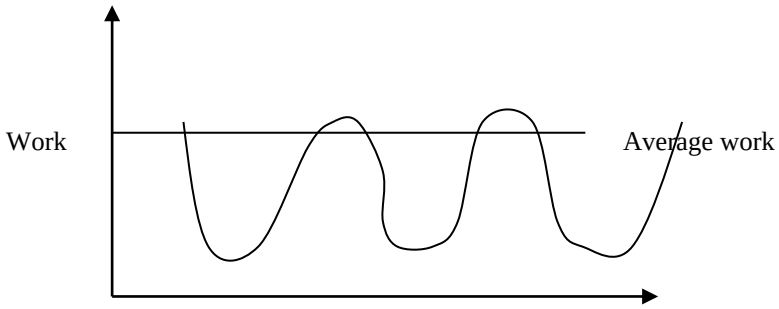
QUANTITATIVE METHODS (FBA 410)

ASSEMBLY LINES BALANCING

Line Balancing could be seen as a solution technique designed to equalise performance times at all work stations of an assembly line. An assembly line, ideally, should have equal work content (measured in terms of time) assigned to them. However, contrary is often the case, due to imbalances in most assemblies' lines of many production systems (unequal work content).

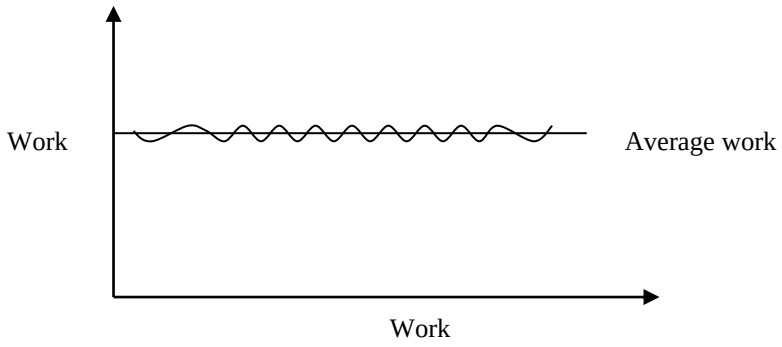
Therefore, assembly line balancing is out to allocate equal work content or task to equal work stations, so as to minimise total idle time of all resources (man, material and machines) by grouping of the various activities subject to precedent requirement.

The following diagrams represent examples of unbalanced and a fairly balanced assembly line:



Work Stations

An unbalanced Assembly line



QUESTION

9.7 The table below shows the precedence relationship of the various tasks of an assembly line. If the desired daily output is 30 units and the daily working hours is 8 hours:

Assembly Tasks (i)	**Task Description**	**Task Time (in mins)**	**Precedence Relationship**
A	Boil	9	-
B	Steer	5	-
C	Filter	4	-
D	Colour	5	A,B
E	Spray	6	D
F	Design	5	E,C
G	Dry	7	F
H	Clean	7	G

You are required to:

1. Construct the precedence/activity/Network diagram of the tasks.
2. Select an appropriate heuristic method for allocating the tasks.
3. Compute the cycle time.
4. Compute the ideal numbers of work stations.
5. Calculated the total activities time and total idle time.
6. Computer the balance delays (BD) of the line (efficiency of the line).

The steps or mechanics of solving line balancing problem, using wester and kilbridge's heuristic method are:

1. Define the various tasks or activities involved in getting the process completed.
2. Draw the Precedence/activity diagram for the process or job.
3. Calculate the available cycle time (C);

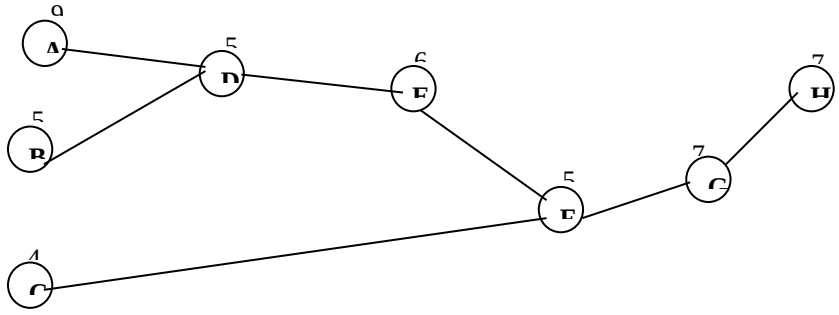
   $$= \frac{\text{Working time per day}}{\text{Desired number of outputs/day}}$$

4. Derive and compute the minimum numbers of work stations

   "N" $= \frac{\sum ti}{C}$
5. Allocate the tasks to the work stations. *Note that* the task to each work station must not be greater than the cycle time "C".

Calculate the efficiency of the system, by calculating the balance delay "BD" = $NC - \frac{\sum ti}{NC} \times 100\%$

1. The precedence diagram of the task:



2. The heuristic method: Wester and Kilbridge
3. The cycle time "C":

$$C = \frac{\text{Available working time per day}}{\text{Desired no of output per day}}$$

$$C = \frac{8 \times 60}{30} = 16 \text{ mins}$$

4. Numbers of appropriate work station "N" $= \frac{\sum ti}{C}$

$$= \frac{48}{16} = 3 \qquad \therefore \quad N = 3$$

Since the appropriate work stations $= 3$

$\therefore$ The appropriate precedence relationship table is as follows:

Work Station	**Task (ti)**	**Duration in Minutes**	**EXCESS TIME**
1	A,B	14	2 Mins
2	C,D,E	15	1 Mins
3	F,G	12	4 Mins
TOTAL	**$\sum$ti**	**41**	***7 Mins***

5. ***$\therefore$ Total activity time (ti) = 41 mins***
Total idle time = 7 mins
Idle time can also be calculated this way: NC - $\sum$ti
Where: N = 3
C = 16 mins
$\sum$ti = 41 mins
$\therefore$ Idle time = 3(16) - 41 = 7 Min

6. The Balance delay (BD) of the line or the efficiency of the line:
Note that: the smaller the balance delays, the more preferred the line is.

$$\therefore \text{ BD} = \frac{NC - \sum ti}{NC} \times 100\%$$

$$= \frac{3(16) - 41}{3(16)} \times 100\%$$

$$= \frac{48 - 41}{} \times 100\%$$

$$= \frac{7}{48} \times 100\%$$

∴ BD = <u>14.58%</u>